

Macroeconomics #7

Dynamic General Equilibrium and Real Business Cycles

Quang-Thanh Tran

RCPD, Tohoku University

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Today's lecture

- ① Intro
- ② Model
- ③ Steady state
- ④ Log-linearization
- ⑤ Some analyses

What is the RBC Model?

The Real Business Cycle (RBC) model is the neoclassical growth model augmented with:

- **Variable labor supply** — households choose how much to work
- **Stochastic productivity shocks** — the driving force of business cycles

Key references:

- Kydland and Prescott (1982) — “Time to Build and Aggregate Fluctuations”
- King, Plosser, and Rebelo (1988)
- King and Rebelo (2000)

Central question: Can a simple, frictionless model driven only by technology shocks replicate the key features of business cycle data?

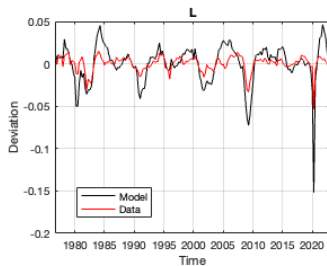
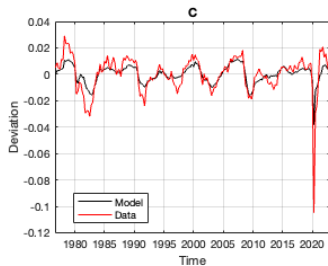
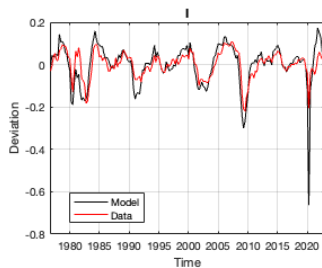
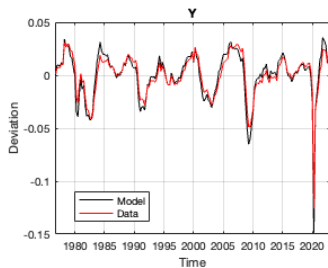


Figure: RBC versus Postwar US data

Building and analyzing the RBC model involves several steps:

- 1 **Build the model:** Specify the model's components, derive optimal solutions for each party
- 2 **Derive key equations:** general equilibrium
- 3 **Solve the model:** Log-linearization, then solve the state-space representation.
- 4 **Calibrate/Estimate:** assign parameter values using data targets, find steady states
- 5 **Evaluate:** compare model moments and impulse responses with data

Model

- Closed economy with a representative household and a representative firm
- Firm uses labor N_t and capital K_t to produce output Y_t
- Household supplies labor and capital, chooses consumption and leisure
- All values are real (no nominal frictions)

Two equivalent formulations:

- ① Firms own capital; households hold bonds and receive dividends
- ② Households own capital and rent it to firms

Both yield identical equilibrium conditions (Modigliani–Miller).

Production Technology

The representative firm produces output using a CRS production function:

$$Y_t = A_t F(K_t, N_t)$$

where A_t is total factor productivity (TFP).

The production function satisfies:

- Essentiality: $F(0, N) = F(K, 0) = 0$
- Strict monotonicity and concavity:
 $F_K > 0, F_{KK} < 0, F_N > 0, F_{NN} < 0$
- Inada conditions: $\lim_{K \rightarrow 0^+} F_K = +\infty, \lim_{K \rightarrow \infty} F_K = 0$

TFP follows an AR(1) process in logs:

$$\ln A_t = \rho \ln A_{t-1} + s_A \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \quad |\rho| < 1$$

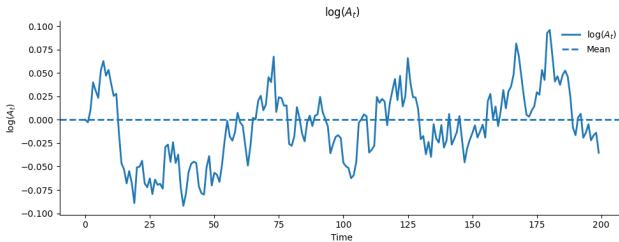


Figure: log form

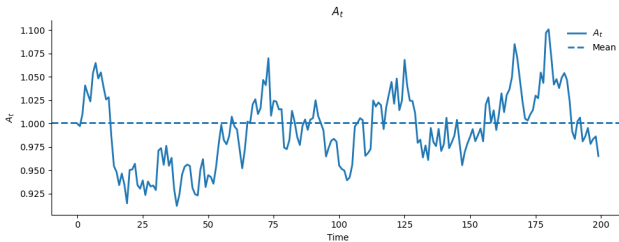


Figure: level

Firm's Profit Maximization

The firm maximizes profit $\Pi_t = A_t F(K_t, N_t) - w_t N_t - R_t^k K_t$.

First-order conditions (factor demands):

$$w_t = A_t F_N(K_t, N_t) \quad (1)$$

$$R_t^k = A_t F_K(K_t, N_t) \quad (2)$$

With Cobb–Douglas production $F(K, N) = K^\alpha N^{1-\alpha}$:

$$w_t = (1 - \alpha) A_t K_t^\alpha N_t^{-\alpha}$$

$$R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$$

Household

A representative household maximizes expected lifetime utility:

$$\max_{C_t, N_t, K_{t+1}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - N_t)$$

where $\beta \in (0, 1)$ is the discount factor, C_t is consumption, N_t is labor, and $1 - N_t$ is leisure (time endowment normalized to 1).

The budget constraint (households own capital):

$$C_t + K_{t+1} - (1 - \delta)K_t = w_t N_t + R_t^k K_t$$

or equivalently:

$$C_t + I_t = w_t N_t + R_t^k K_t$$

with the law of motion for capital:

$$K_{t+1} = I_t + (1 - \delta)K_t$$

Household's First-Order Conditions

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(C_t, 1 - N_t) + \lambda_t [w_t N_t + R_t^k K_t + (1 - \delta)K_t - C_t - K_{t+1}] \right\}$$

FOC

$$(C_t) : \quad \frac{\partial \mathcal{L}}{\partial C_t} = u_C - \lambda_t = 0,$$

$$(N_t) : \quad \frac{\partial \mathcal{L}}{\partial N_t} = -u_N + \lambda_t w_t = 0,$$

$$(K_{t+1}) : \quad \frac{\partial \mathcal{L}}{\partial K_{t+1}} = -\lambda_t + \lambda_{t+1} [R_{t+1}^k + (1 - \delta)] = 0,$$

Envelope theorem implies

$$\begin{aligned} u_C(C_t, 1 - N_t) &= \lambda_t, \\ u_C(C_{t+1}, 1 - N_{t+1}) &= \lambda_{t+1} \end{aligned}$$

Consumption–leisure trade-off:

$$\frac{u_l(C_t, 1 - N_t)}{u_C(C_t, 1 - N_t)} = w_t \quad (3)$$

The marginal rate of substitution between leisure and consumption equals the real wage.

Euler equation for capital:

$$u_C(C_t, 1 - N_t) = \beta \left[u_C(C_{t+1}, 1 - N_{t+1}) (R_{t+1}^k + 1 - \delta) \right] \quad (4)$$

The marginal utility of consuming today equals the expected discounted marginal utility from saving one more unit, earning the return $R_{t+1}^k + 1 - \delta$, and consuming tomorrow.

Competitive Equilibrium

Definition. Given K_0 and $\{A_t\}_{t=0}^{\infty}$, a competitive equilibrium is a set of prices $\{w_t, R_t^k\}$ and allocations $\{C_t, N_t, K_{t+1}, Y_t, I_t\}$ such that:

- 1 Households optimize (Euler equation + labor supply)
- 2 Firms optimize (factor demand conditions)
- 3 Goods market clears: $Y_t = C_t + I_t$

The **equilibrium system** (8 equations, 8 unknowns):

- | | |
|------------------------|-----------------------|
| 1 Euler equation | 5 Production function |
| 2 Labor supply | 6 Resource constraint |
| 3 Capital accumulation | 7 Wage equation |
| 4 TFP process | 8 Capital rental rate |

Preferences:

$$u(C_t, N_t) = \frac{C_t^{1-\eta}}{1-\eta} - \theta \frac{N_t^{1+\varphi}}{1+\varphi}$$

- $\eta > 0$: inverse of the intertemporal elasticity of substitution (IES = $1/\eta$)
- $\varphi \geq 0$: inverse of the Frisch labor supply elasticity
- $\theta > 0$: weight on labor disutility
- When $\eta = 1$: $u(C) = \ln C$ (log utility)

FOC

$$\frac{u_l}{u_C} = \frac{\theta N_t^\varphi}{C_t^{-\eta}} = \theta C_t^\eta N_t^\varphi = w_t$$

Consumption

- High η , low IES, consumption is less responsive to change in interest, households prefer consumption smoothing
- Low η , high IES, consumption is more responsive to change in interest, households are more flexible in changing consumption

Labor

- High φ , Frisch elasticity $1/\varphi$ is low, labor is less elastic, workers dislike additional work, so they are less responsive to wage changes
- Low φ , Frisch elasticity $1/\varphi$ is high, labor is more elastic, workers adjust hours easily, so they are more responsive to wage changes

Production:

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}, \quad \alpha \in (0, 1)$$

Factor prices

$$w_t = (1 - \alpha) A_t K_t^\alpha N_t^{-\alpha}$$

$$R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$$

Goods market clear

$$Y_t = C_t + I_t$$

Equilibrium Conditions with Functional Forms

Substituting the functional forms, the equilibrium system becomes:

$$\text{Euler: } C_t^{-\eta} = \beta \mathbb{E}_t \left[C_{t+1}^{-\eta} (R_{t+1}^k + 1 - \delta) \right]$$

$$\text{Labor supply: } \theta C_t^\eta N_t^\varphi = w_t$$

$$\text{Capital accumulation: } K_{t+1} = (1 - \delta)K_t + I_t$$

$$\text{Production: } Y_t = A_t K_t^\alpha N_t^{1-\alpha}$$

$$\text{Capital demand: } R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$$

$$\text{Labor demand: } w_t = (1 - \alpha) A_t K_t^\alpha N_t^{-\alpha}$$

$$\text{Resource constraint: } Y_t = C_t + I_t$$

$$\text{TFP process: } \ln A_t = \rho \ln A_{t-1} + s_A \varepsilon_t$$

Steady State

Non-Stochastic Steady State (I)

In the steady state: $K_{t+1} = K_t = K$, $C_{t+1} = C_t = C$, $A = 1$, $\varepsilon = 0$.

Step 1: From the Euler equation (uncertainty vanishes):

$$R^k = \frac{1}{\beta} - 1 + \delta$$

The steady-state rental rate is pinned down by preferences and depreciation alone.

Step 2: From capital demand $R^k = \alpha(K/N)^{\alpha-1}$:

$$\frac{K}{N} = \left(\frac{\alpha}{R^k} \right)^{\frac{1}{1-\alpha}}$$

Step 3: Steady-state wage:

$$w = (1 - \alpha) \left(\frac{K}{N} \right)^{\alpha}$$

Non-Stochastic Steady State (II)

Step 4: From capital accumulation, $I = \delta K$, so:

$$\frac{I}{N} = \delta \frac{K}{N}$$

Step 5: From the resource constraint:

$$\frac{C}{N} = \left(\frac{K}{N}\right)^\alpha - \delta \frac{K}{N}$$

Step 6: Set target $N = 1/3$ (empirical target), then back out θ :

$$\theta = \frac{w}{C^\eta N^\varphi}$$

With N known, all steady-state values (K, Y, C, I, w, R^k) follow suit.

Key ratios: $I/Y = \alpha\delta/(1/\beta - 1 + \delta) \approx 0.225$, $C/Y \approx 0.775$.

Log-linearization

Why Log-Linearize?

The equilibrium system is a set of **nonlinear stochastic difference equations**. No closed-form solution exists in general.

Strategy: Approximate each equation around the non-stochastic steady state using a first-order Taylor expansion in log-deviations.

Define the log-deviation of variable X_t :

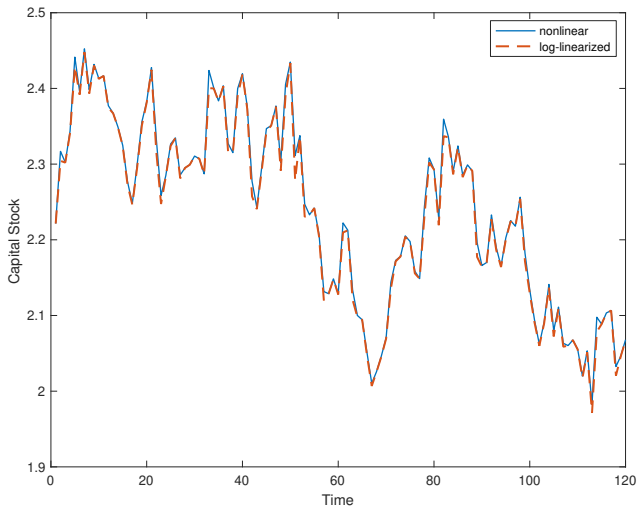
$$\tilde{X}_t = \ln X_t - \ln \bar{X} \approx \frac{X_t - \bar{X}}{\bar{X}}$$

Uhlig's method: substitute $X_t = \bar{X} e^{\tilde{X}_t}$ and use the approximation rules:

$$e^{\tilde{X}_t} \approx 1 + \tilde{X}_t$$

$$e^{\tilde{X}_t + a\tilde{Y}_t} \approx 1 + \tilde{X}_t + a\tilde{Y}_t$$

$$\tilde{X}_t \cdot \tilde{Y}_t \approx 0$$



Linearized Equilibrium System (I)

Applying Uhlig's method to each equation:

Consumption–leisure:

$$\eta \tilde{C}_t + \varphi \tilde{N}_t = \tilde{w}_t$$

Euler equation:

$$\frac{\eta}{\beta} \mathbb{E}_t(\tilde{C}_{t+1} - \tilde{C}_t) = R^k \mathbb{E}_t \tilde{R}_{t+1}^k$$

Law of motion for capital:

$$\tilde{K}_{t+1} = (1 - \delta) \tilde{K}_t + \delta \tilde{I}_t$$

Production:

$$\tilde{Y}_t = \tilde{A}_t + \alpha \tilde{K}_t + (1 - \alpha) \tilde{N}_t$$

Linearized Equilibrium System (II)

Capital demand:

$$\tilde{R}_t^k = \tilde{Y}_t - \tilde{K}_t$$

Labor demand:

$$\tilde{w}_t = \tilde{Y}_t - \tilde{N}_t$$

Resource constraint:

$$\bar{Y} \tilde{Y}_t = \bar{C} \tilde{C}_t + \bar{I} \tilde{I}_t$$

TFP process (already log-linear):

$$\tilde{A}_t = \rho \tilde{A}_{t-1} + s_A \varepsilon_t$$

This is a system of **8 linear equations** in 8 log-deviation variables:
 $\tilde{Y}_t, \tilde{C}_t, \tilde{N}_t, \tilde{w}_t, \tilde{R}_t^k, \tilde{I}_t, \tilde{K}_t, \tilde{A}_t$.

Reducing to a Three-Equation System

By substituting the static equations, we can reduce the system to 3 dynamic equations in $(\widetilde{K}_t, \widetilde{A}_t, \widetilde{C}_t)$.

Output as a function of states and consumption:

$$\widetilde{Y}_t = \psi_1 \widetilde{A}_t + \psi_2 \widetilde{K}_t - \psi_3 \widetilde{C}_t$$

where

$$\psi_1 = \frac{1 + \varphi}{\alpha + \varphi}, \quad \psi_2 = \frac{\alpha(1 + \varphi)}{\alpha + \varphi}, \quad \psi_3 = \frac{(1 - \alpha)\eta}{\alpha + \varphi}$$

Capital accumulation:

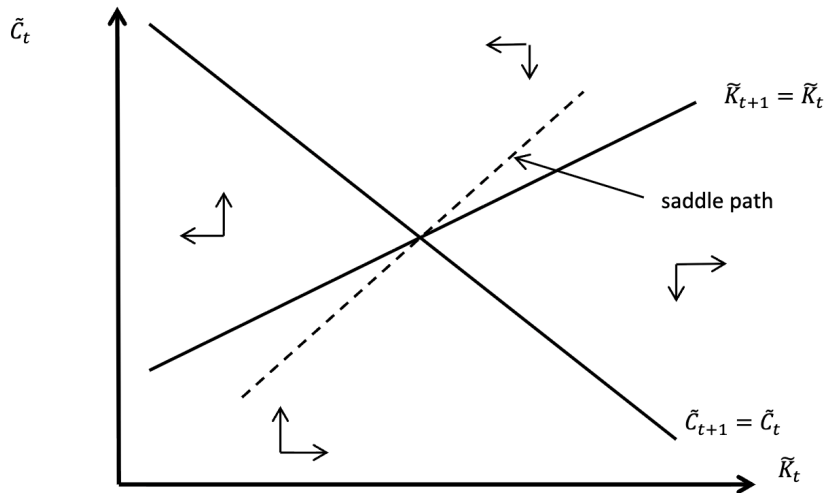
$$\widetilde{K}_{t+1} = \gamma_k \widetilde{K}_t + \gamma_A \widetilde{A}_t - \mu \widetilde{C}_t$$

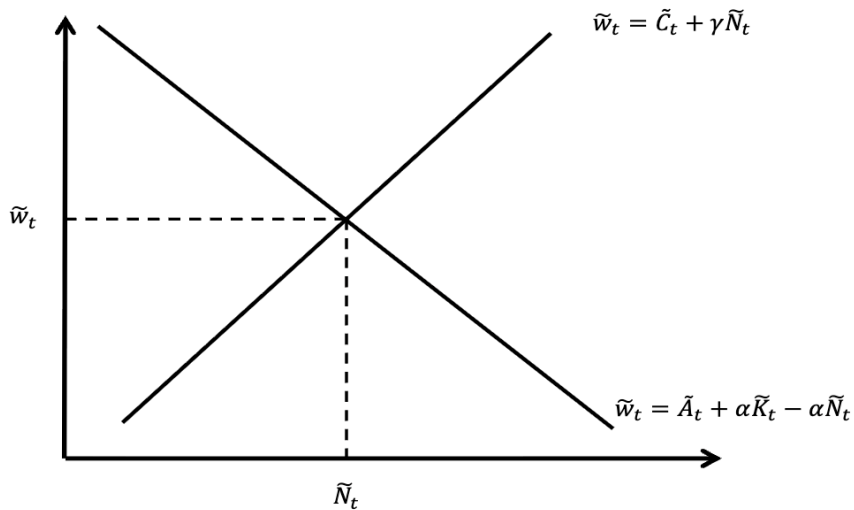
Euler equation: involves $\mathbb{E}_t \widetilde{C}_{t+1}$ and $\mathbb{E}_t \widetilde{R}_{t+1}^k$. After some lengthy derivations

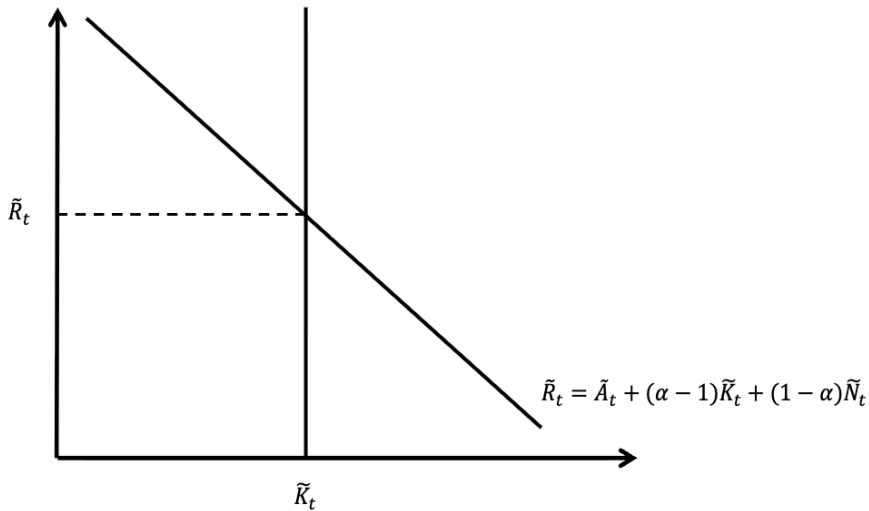
$$\widetilde{C}_t = \frac{\theta + \varphi}{1 - \alpha} \widetilde{A}_t - \varphi \widetilde{K}_t$$

Analysis

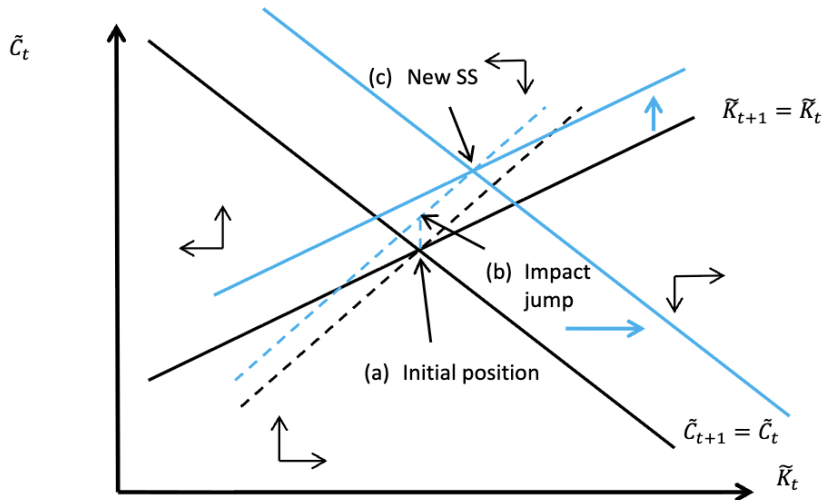
Saddle path Equilibrium

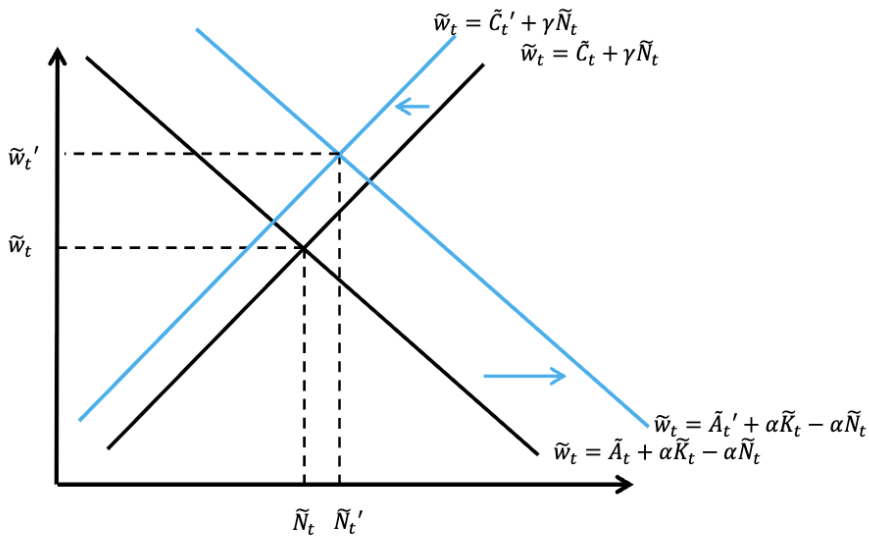


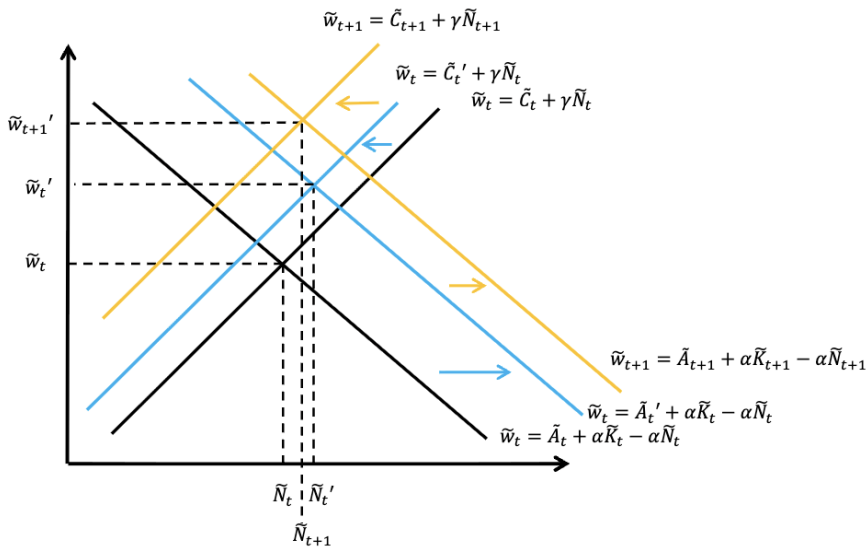




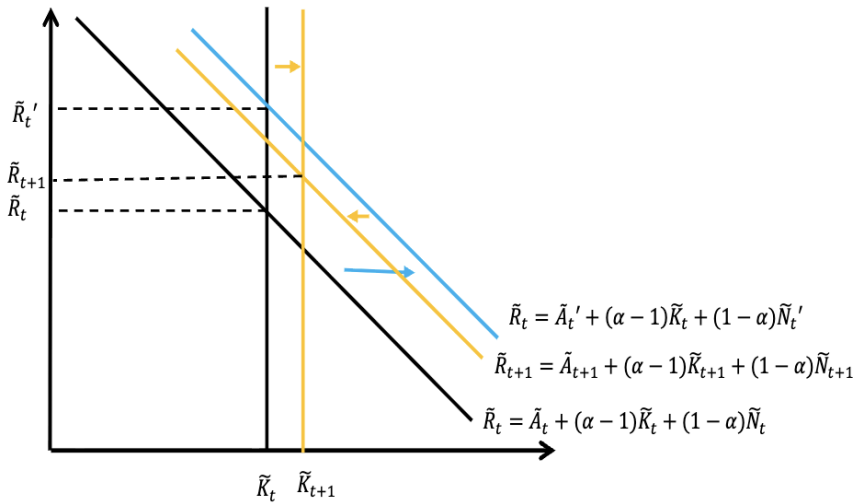
Positive Productivity Shock











Permanent Productivity Shock

Suppose $\tilde{A}_t$ jumps up permanently. Both isoclines shift up.

Impact effects:

- Consumption jumps up (income effect dominates for most parameterizations)
- Investment rises (need to build capital toward higher steady state)
- Output rises: directly through A_t and through higher N_t
- Real wage rises unambiguously
- Labor: ambiguous (substitution vs. income effect), likely rises
- R_t^k rises on impact

Transition dynamics:

- C_t and K_t gradually increase toward new (higher) steady state
- Investment overshoots initially, then settles at higher level
- R_t^k returns to original steady state: $R^k = 1/\beta - 1 + \delta$
- Labor may initially rise then fall back (with KPR preferences)

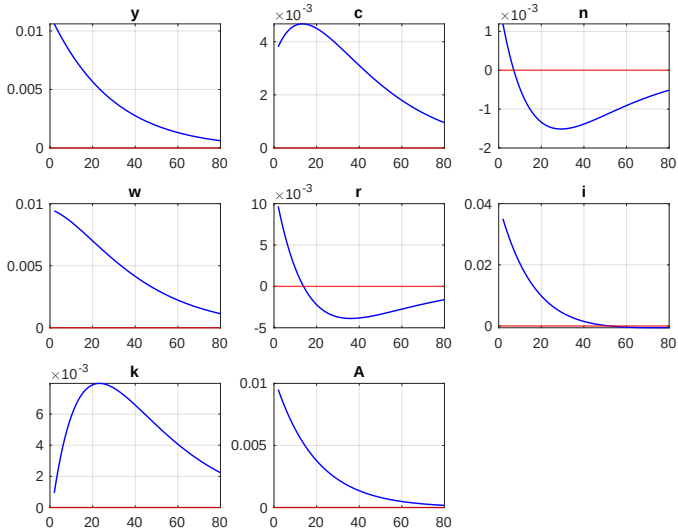
Temporary Productivity Shock

A purely temporary (iid) shock to $\tilde{A}_t$: lasts only one period.

Key differences from permanent shock:

- Consumption barely reacts (permanent income logic)
- **Labor rises more** (smaller income effect $\Rightarrow$ substitution effect dominates)
- **Output rises more** in relative terms (stronger labor amplification)
- Real wage rises less
- R_t^k rises on impact, quickly returns
- All variables return to steady state quickly

General principle: More persistent shocks $\Rightarrow$ responses look more like permanent case. More transitory $\Rightarrow$ responses look more like iid case. Standard calibration ($\rho \approx 0.98$) falls between the two.



Idea: Choose parameters so the model's **steady state** matches long-run features of the data. Then use the *same* model to explain short-run fluctuations.

Parameter	Target / Source	Value
α	Capital's share of income	1/3
β	Average real interest rate $\approx 2\%/yr$	0.99 (quarterly)
δ	Investment–output ratio $\approx 22.5\%$	0.02 (quarterly)
θ	Steady-state $N = 1/3$	≈ 1.71
ρ	AR(1) of linearly detrended TFP	0.979
s_A	Std. dev. of TFP innovations	0.009

Stylized US Business Cycle Facts

HP-filtered US data (1947Q1–2022Q3), $\lambda = 1600$:

Series	Std. Dev.	Rel. Std. Dev.	Corr w/ Y_t
Output	0.0166	1.00	1.00
Consumption	0.0110	0.66	0.76
Investment	0.0444	2.67	0.76
Hours	0.0211	1.27	0.87
Avg. Labor Productivity	0.0114	0.69	0.27
Wage	0.0115	0.69	-0.01
Real Rate	0.0043	0.26	-0.00
TFP	0.0127	0.76	0.78

Key facts: consumption is smoother than output, investment is $\sim 2.5\times$ more volatile, hours are procyclical, all series are persistent.

Model vs. Data

Series	Rel. Std. Dev.		Corr w/ Y_t	
	Data	Model	Data	Model
Consumption	0.66	0.38	0.76	0.96
Investment	2.67	3.25	0.76	0.99
Hours	1.27	0.43	0.87	0.98
Avg. Labor Prod.	0.69	0.58	0.27	0.99
Wage	0.69	0.58	-0.01	0.99
Real Rate	0.26	0.03	-0.00	0.97
TFP	0.76	0.71	0.78	1.00

Successes: output volatility, relative volatility of consumption and investment, persistence, procyclicality of main aggregates.

Failures: hours too smooth, correlations too high, wage and labor productivity perfectly correlated (data: nearly zero).

Is TFP an Accurate Measure of A_t ?

Measured TFP is constructed as a Solow residual:

$$\ln \hat{A}_t = \ln Y_t - \alpha \ln K_t - (1 - \alpha) \ln N_t$$

But if the true production function involves **variable utilization**:

$$Y_t = A_t(u_t K_t)^\alpha (e_t N_t)^{1-\alpha}$$

then: $\ln \hat{A}_t = \ln A_t + \alpha \ln u_t + (1 - \alpha) \ln e_t$.

Measured TFP conflates *true* technology with utilization changes.

Fernald's corrected TFP: after adjusting for utilization, TFP is actually **countercyclical** (correlation with output ≈ -0.21). If true A_t is not procyclical, the RBC model loses its main driving force.

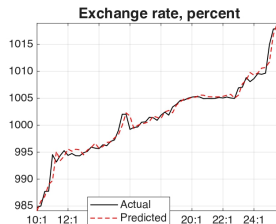
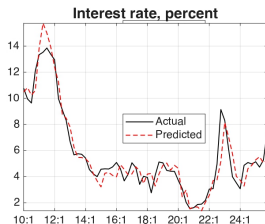
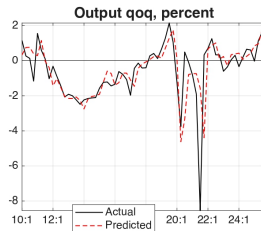
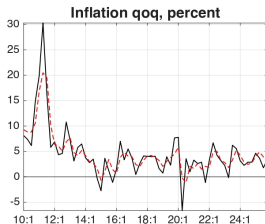
Where Do We Go From Here?

The RBC model is the **backbone** of modern DSGE models.

Extensions include:

- **New Keynesian models:** add sticky prices/wages $\Rightarrow$ monetary policy matters
- **Multiple shocks:** government spending, investment-specific technology, preference shocks
- **Labor market frictions:** search and matching (Mortensen–Pissarides)
- **Financial frictions:** Bernanke–Gertler–Gilchrist financial accelerator
- **Variable capital utilization:** helps with amplification
- **Habit formation:** generates consumption persistence

The methodology (DSGE modeling, log-linearization, calibration, impulse responses) remains the standard approach in quantitative macroeconomics.



Appendix

Appendix: Deriving the Reduced-Form Coefficients

Combining consumption–leisure and labor demand:

$$\tilde{N}_t = \frac{\tilde{Y}_t - \eta \tilde{C}_t}{1 + \varphi}$$

Substituting into the production function $\tilde{Y}_t = \tilde{A}_t + \alpha \tilde{K}_t + (1 - \alpha) \tilde{N}_t$:

$$\tilde{Y}_t = \frac{1 + \varphi}{\alpha + \varphi} \tilde{A}_t + \frac{\alpha(1 + \varphi)}{\alpha + \varphi} \tilde{K}_t - \frac{(1 - \alpha)\eta}{\alpha + \varphi} \tilde{C}_t$$

For the capital accumulation, using $I/K = \delta$ in steady state:

$$\tilde{K}_{t+1} = \underbrace{\left(1 - \delta + \frac{Y}{K} \psi_2\right)}_{\gamma_k} \tilde{K}_t + \underbrace{\frac{Y}{K} \psi_1}_{\gamma_A} \tilde{A}_t - \underbrace{\left(\frac{Y}{K} \psi_3 + \frac{C}{K}\right)}_{\mu} \tilde{C}_t$$

Appendix: Constructing the System Matrices

See **The ABC of RBCs** by George McCandless (2008) for more detail.

The coefficient matrices for the undetermined coefficients method:

$$A = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} -(1 - \delta) \\ -\alpha \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\delta \\ 1 & 0 & -(1 - \alpha) & 0 & 0 & 0 \\ 0 & \eta & \varphi & -1 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ -\bar{Y} & \bar{C} & 0 & 0 & 0 & \bar{I} \end{pmatrix}$$

State-Space Representation

Classify variables:

- **Endogenous state** (predetermined): $x_t = [\widetilde{K}_t]$
- **Exogenous state**: $z_t = [\widetilde{A}_t]$
- **Jump variables**: $y_t = [\widetilde{Y}_t, \widetilde{C}_t, \widetilde{N}_t, \widetilde{w}_t, \widetilde{R}_t^k, \widetilde{I}_t]'$

The linearized system can be written as:

$$\text{Static block: } 0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$\text{Forward block: } 0 = \mathbb{E}_t[Fx_{t+1} + Gx_t + Hx_{t-1} + \dots \\ + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t]$$

$$\text{Exogenous: } z_{t+1} = Nz_t + \varepsilon_{t+1}$$

The **solution** takes the form:

$$\boxed{x_t = Px_{t-1} + Qz_t, \quad y_t = Rx_{t-1} + Sz_t}$$

Method of Undetermined Coefficients

Idea: Guess the form of the solution, substitute into the equilibrium conditions, and match coefficients.

Substituting the guessed policy functions into the system yields:

$$R = -C^{-1}(AP + B)$$

$$S = -C^{-1}(AQ + D)$$

P solves a **quadratic equation**:

$$(F - JC^{-1}A)P^2 - (JC^{-1}B - G + KC^{-1}A)P - KC^{-1}B + H = 0$$

Select the root satisfying $|P| < 1$ (stability).

Q is solved from a linear system using the Kronecker product / vec operator.

Alternative approach: Write the system as

$$B \begin{pmatrix} x_{t+1} \\ \mathbb{E}_t y_{t+1} \end{pmatrix} = A \begin{pmatrix} x_t \\ y_t \end{pmatrix} + G \varepsilon_t$$

Steps:

- ① Compute $Z = B^{-1}A$, find eigenvalues and eigenvectors
- ② Order eigenvalues: stable ($|\lambda| < 1$) first, unstable ($|\lambda| > 1$) last
- ③ **BK condition:** # unstable eigenvalues = # jump variables
- ④ Jump variables: $y_t = -\widehat{M}_{22}^{-1} \widehat{M}_{21} x_t$ (deterministic part)
- ⑤ State transition:
$$x_{t+1} = [\widehat{M}_{11} - \widehat{M}_{12} \widehat{M}_{22}^{-1} \widehat{M}_{21}]^{-1} \bar{\Lambda}_{11} [\widehat{M}_{11} - \widehat{M}_{12} \widehat{M}_{22}^{-1} \widehat{M}_{21}] x_t$$

Both methods produce **numerically identical** results.